Positive Square Energy for Rank-Seven Kernels of Order Eight

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Abstract

Let G be a finite simple connected graph of cyclomatic rank seven whose block decomposition has exactly one positive-rank cyclic block. Suppose that, after suppressing the degree-two vertices of that block, its loopless 2-connected multigraph kernel has eight vertices. We prove $s^+(G) \geq |V(G)|$, where s^+ is the sum of the squares of the positive adjacency eigenvalues. Recent general theorems already cover every tree-free member of this class; the additional scope here is the uniform allowance of arbitrary rooted-tree attachments, including leaf-bearing instances. The proof combines an exact census of the 4,015 order-eight kernels and their 26,426,026 physical parity orbits with analytic coarse certificates and a disjoint exact certificate partition of the 493,417 residual orbits. The last residual is handled by an induced partition into two tricyclic territories. The executable theorem gate pins every proof source and artifact and offers a quick integrity mode and a full read-only replay mode. Three expensive historical base-producer outputs remain explicit immutable trust roots; full mode freshly exact-replays the 29,000 direct-rational witness records rather than retaining them as a fourth root. No claim is made for kernel orders nine through twelve, for multiblock rank-seven graphs, or for all connected heptacyclic graphs.

1 Statement, scope, and spectral tools

All graphs in this paper are finite, simple, and undirected. If the adjacency eigenvalues of a graph H are $\lambda_1, \ldots, \lambda_n$, write

$$s^+(H) = \sum_{\lambda_i > 0} \lambda_i^2, \qquad s^-(H) = \sum_{\lambda_i < 0} \lambda_i^2.$$

A connected graph has cyclomatic rank $\beta(H) = |E(H)| - |V(H)| + 1$. A *cyclic block* means a maximal 2-connected block of positive cyclomatic rank; bridges are rank-zero blocks.

Theorem 1.1 (Scoped order-eight kernel theorem)**.** *Let G be a finite simple connected graph with $\beta(G) = 7$. Assume that G has exactly one positive-rank cyclic block B. Suppress every degree-two vertex of B, and let K be the resulting loopless 2-connected multigraph. If $|V(K)| = 8$, then*

$$\boxed{s^+(G) \geq |V(G)|}.$$

Equivalently, the conclusion holds for every simple subdivision of every rank-seven order-eight kernel, with arbitrary finite rooted trees attached at branch vertices or subdivision vertices.

The hypotheses in Theorem 1.1 are indispensable to the scope of this manuscript. They do not include rank-seven graphs with two or more positive-rank blocks, and they do not include kernels of orders nine through twelve. In particular, the theorem is not a theorem about all connected graphs with $|E| = |V| + 6$.

Relation to the square-energy literature. Akbari, Kumar, Mohar, Pragada, and Zhang conjectured that every connected graph of order n and size $m \geq n+1$ satisfies $s^+(G) \geq n$ [1]. A connected graph of cyclomatic rank seven has $m = n+6$, so Theorem 1.1 verifies only the restricted class stated above. Akbari, Hu, and Liu subsequently proved the conclusion for every 2-connected noncycle [2] and, more generally, for every connected noncycle of minimum degree at least two [3]. Consequently every tree-free member of our class is already covered by those results. Within the one-cyclic-block class considered here, any nontrivial finite rooted-tree attachment has a non-root leaf, so the resulting graph has minimum degree one and is outside the latter theorem. The certificate argument below treats all such attachments uniformly. The general $n-1$ square-energy theorem [4] gives a baseline one unit below the claim here. We make no claim of firstness: the dated search audit [9] located no previous theorem with the full combined scope, but a negative search does not establish novelty or priority.

We use two standard spectral devices. The first is the induced partition inequality established in the square- and p-energy literature [6, 5]: positive square energy is superadditive over induced vertex territories.

Lemma 1.2 (Induced superadditivity)**.** *If $V(H) = V_1 \dot\cup \cdots \dot\cup V_r$, then*

$$s^+(H) \geq \sum_{j=1}^{r} s^+(H[V_j]).$$

Proof. For a real symmetric matrix A,

$$\operatorname{tr}(A_+^2) = \max_{X \succeq 0} \big(2\operatorname{tr}(AX) - \operatorname{tr}(X^2)\big).$$

For $A = A(H)$, test this variational formula with the block-diagonal matrix whose blocks are $A(H[V_j])_+$. Cross-territory entries of A do not contribute, and the displayed inequality follows. □

Second, for an edge-containing graph H, define the DNN constant

$$\kappa(H) = \min_{\substack{R \succeq 0 \\ R_{vv}=1}} \sum_{uv \in E(H)} \frac{2}{1 - R_{uv}}, \tag{1}$$

with boundary values interpreted by limits. The correlation dual for the optimized doubly nonnegative bound gives

$$s^-(H) \leq \kappa(H). \tag{2}$$

Moreover, κ is additive under one-vertex sums and $\kappa(T) = |E(T)|$ for every tree T; proofs are recorded in the companion tricyclic manuscript [10]. These statements may also be obtained from the primal DNN formulation by orthogonally gluing the Gram data at a cut vertex.

2 Suppression and the finite frontier

Let B be a 2-connected graph of cyclomatic rank seven. Suppressing its degree-two vertices gives a loopless 2-connected multigraph K of minimum degree at least three, and

$$|E(K)| = |V(K)| + 6, \qquad 3|V(K)| \leq 2|E(K)|.$$

Consequently $2 \leq |V(K)| \leq 12$. Conversely, B is obtained by replacing each physical edge of K by a positive-length path, with pairwise disjoint interiors. Simplicity permits at most one unit path

in each parallel class. Every part of G outside B is a rooted tree attached at one vertex of B. The complete removable-ear generation theorem and its exact audit give 4,015 order-eight kernels [11].

Fix a rank-seven kernel K of order eight. It has fourteen physical edges. For a parallel class of multiplicity m containing o odd replacement paths, define its canonical simple lengths, up to permutation within the class, by

$$o = 0 : \ (2, \dots, 2), \qquad o > 0 : \ (1, 3, \dots, 3, 2, \dots, 2). \tag{3}$$

Thus every simple realization in the same physical parity orbit can be ordered as $\ell = c + 2d$ with $d \geq 0$ coordinatewise. For a canonical vector $c = (c_1, \dots, c_{14})$, put

$$\mathcal{F}(c) = \{c\} \cup \{c + 2e_i : 1 \leq i \leq 14\}. \tag{4}$$

The exact path atom explains both the DNN budget and the all-length lift.

Lemma 2.1 (Exact path elimination)**.** *Fix unit branch vectors with correlation r at the endpoints of a path of length $q \geq 1$. Its least excess over the baseline contribution q in (1) is*

$$f_q(r) = q \tan^2\left(\frac{\arccos((-1)^q r)}{2q}\right). \tag{5}$$

For fixed r and fixed parity, $f_{q+2}(r) \leq f_q(r)$, with strict inequality unless the transformed endpoint angle vanishes.

Proof. Alternately negate the vectors along the path. The transformed endpoints have angular distance $\beta = \arccos((-1)^q r)$. The spherical triangle inequality and convexity of $\tan^2(x/2)$ show that q equal angular steps minimize the sum, yielding (5). For real $q > 0$, differentiation of $q \tan^2(\beta/(2q))$ has the sign of $\sin z \cos z - 2z$, where $z = \beta/(2q)$; this is negative for $z > 0$. Endpoint cases follow by limits. □

Suppose a retained branch Gram has total path excess at most six. If its subdivision core has L edges, then $\kappa(B) \leq L + 6$. If t tree edges are attached, one-vertex additivity gives $\kappa(G) \leq L + 6 + t$. Since rank seven implies $|V(B)| = L - 6$, equations (2) and $s^+(G) + s^-(G) = 2|E(G)|$ give

$$s^+(G) \geq 2(L + t) - (L + 6 + t) = L - 6 + t = |V(G)|. \tag{6}$$

Lemma 2.1 promotes a canonical or coordinate-frontier Gram to every same-parity realization that dominates it. A structural owner is admitted only when it supplies its own all-length and rooted-tree argument.

3 Exact census and owner partition

The order-eight census enumerates all physical parity rows and quotients them by the full multigraph automorphism group. It independently checks every materialized orbit count with Burnside's lemma. The resulting totals are

quantity	exact count
order-eight kernels	4,015
physical parity rows	33,006,092
automorphism orbits	26,426,026
coarse-certified orbits	25,932,609
coarse residual orbits	493,417
residual targets in (4)	7,401,255

The exact manifest and its eight authenticated chunks satisfy $25{,}932{,}609 + 493{,}417 = 26{,}426{,}026$. The coarse certificate is a four-color regular-tetrahedron Gram whose exact scaled cost is at most the rank-seven excess budget six; hence its accepted rows are already DNN owners. The remaining 493,417 rows are partitioned by the theorem gate as follows.

exclusive owner	residual orbits	targets
payload-free symbolic lanes	605	9,075
direct rational witness packs	29,000	435,000
scalar SOS, after precedence	194,351	2,915,265
typed-diagonal Gram, after precedence	185,309	2,779,635
boundary shared-Gram rescue	296	4,440
structural cycle-Gram lanes I–III	245	3,675
packet lanes IV–XII	29,013	435,195
global post-XII shared packets	54,597	818,955
final induced-territory owner	1	15
exact total	493,417	7,401,255

Table 1: Disjoint precedence partition of the coarse residual frontier.

The row identity in Table 1 is

$$\begin{aligned} 493{,}417 = {} & 605 + 29{,}000 + 194{,}351 + 185{,}309 + 296 \\ & + 245 + 29{,}013 + 54{,}597 + 1. \end{aligned} \tag{7}$$

The first five lanes leave 83,856 rows. Structural lanes I–III leave 83,611. Packet lanes IV–XII leave 54,598, and the global post-XII lane leaves one. Thus the exact remainder chain used by the master gate is

$$83{,}856 \longrightarrow 83{,}611 \longrightarrow 54{,}598 \longrightarrow 1. \tag{8}$$

We summarize why the entries are theorem owners rather than merely successful numerical searches.

- The payload-free lanes use rank-one signings, diagonally dominant signed imbalance Grams, or exact simplex/mixed-atom PSD completions.
- The scalar and typed-diagonal lanes express every accepted Gram as an exact sum of rational Gram squares and a nonnegative diagonal completion. Twenty gap-free typed receipts cover all 492,812 rows after the payload-free lane; the precedence bridge verifies the exact exclusive sets.
- Every direct, rescue, structural, and packet witness is accepted only after exact rational replay. For a shared packet, numerical optimization is proposal-only; acceptance verifies the unit branch Gram and each path-chain cost with rational arithmetic.
- The structural verifier replays all 245 stored owners and all 3,675 frontier costs. The packet-chain verifier replays the nine exact complements IV–XII, whose owner counts sum to 29,013.
- Each DNN or packet owner retains a branch Gram and invokes Lemma 2.1, so every same-parity length vector is covered. New tree vertices can be assigned the negative of their parent vector, adding one to both κ and the edge count. This gives arbitrary rooted-tree attachments.

The immutable trust boundary is important. Three expensive historical producer searches remain roots: (1) payload-free recognizer semantics, (2) raw scalar-SOS membership semantics, and (3) the 492,812 typed searches behind the twenty receipts. The base precedence bridge reports the historical four-root boundary, but the master discharges its fourth item by freshly canonical-decoding and exact-`Fraction` replaying every one of the 29,000 direct-rational witness records, covering 435,000 targets. The full master also reconstructs the selected 492,812-tuple projection in memory from the authenticated census chunks and checks the pinned cache field-for-field; it neither writes a newly regenerated cache artifact nor reruns the three searches just named. Separate read-only verifiers replay the structural and packet witnesses. The final claim is therefore a complete exact certificate chain relative to three explicitly named and byte-pinned base producer outputs, not a claim that every historical search was rerun.

4 The unique residual orbit

After the post-XII replay, the unique residual is the six-field row

$$[416536, 416987, 6832, E, p, 2], \tag{9}$$

where

$$E = \{03, 06, 07, 12, 14, 15, 24, 25, 36, 37, 45, 47, 56, 67\},$$
$$p = (1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1).$$

The first two entries are respectively its stream index and residual source index, 6832 is the global kernel index, and the orbit size is two. Its support consists of two vertex-disjoint copies of K_4, on

$$A = \{0, 3, 6, 7\}, \qquad C = \{1, 2, 4, 5\},$$

joined by the two physical paths P_{74} and P_{65}. The following proposition is stronger than the fifteen finite obligations in (4).

Proposition 4.1 (Two tricyclic territories)**.** *Start with the two displayed copies of K_4, replace all twelve internal K_4 edges and the two connectors $74, 65$ by arbitrary positive-length paths with pairwise disjoint interiors, and attach arbitrary finite rooted trees at branch or subdivision vertices. The resulting graph H satisfies $s^+(H) \geq |V(H)|$, with no parity restriction on any path.*

Proof. Let C_A and C_C be the two K_4 subdivisions. Orient both connector paths from the A side to the C side. Put in U_0 all vertices of C_A and all internal connector vertices, and put in W_0 all vertices of C_C. Every attached rooted tree follows its root, producing a vertex partition $V(H) = U \dot\cup W$.

The induced graph $H[U]$ consists of C_A, the two connector stubs obtained by deleting their terminal C-side branch vertices, and rooted forests meeting this cyclic core in one root. The graph $H[W]$ consists of C_C and its assigned rooted forests. Both induced graphs are connected. This description also covers a unit connector: then its sole edge crosses the partition and it has no internal stub.

If C_A has L_A edges and its retained forest has t_A edges, then a subdivision of K_4 has $L_A - 2$ vertices, whence

$$|E(H[U])| = L_A + t_A = |V(H[U])| + 2.$$

The identical calculation holds for $H[W]$. The tricyclic theorem says that every finite simple connected graph J with $|E(J)| = |V(J)| + 2$ satisfies $s^+(J) \geq |V(J)|$ [10]. Therefore Lemma 1.2 gives

$$s^+(H) \geq s^+(H[U]) + s^+(H[W]) \geq |U| + |W| = |V(H)|.$$

No edge-addition monotonicity, connector debit, or DNN gluing is used. □

The singleton artifact binds (9) byte-for-byte to Proposition 4.1. Because the proposition covers all lengths, all parities, and arbitrary rooted trees, it supplies the last owner in (7).

5 Proof of the theorem

Proof of Theorem 1.1. Suppress B to its rank-seven order-eight kernel K. The exhaustive kernel census and physical automorphism quotient place every parity state of K in exactly one of two sets.

If it is coarse-certified, its tetrahedral correlation Gram has excess at most six and equation (6) applies. Otherwise it is one of the 493,417 residual orbits. The disjoint exact identity (7) assigns it to exactly one row of Table 1. Every owner except the last supplies an exact DNN or packet certificate together with the all-length and rooted-tree lift described above, so again (6) applies. The unique last orbit is covered uniformly by Proposition 4.1. These alternatives exhaust every kernel, physical parity orbit, simple subdivision length, and rooted-tree attachment within the stated scope. Hence $s^+(G) \geq |V(G)|$. □

6 Executable verification and integrity boundary

The authoritative integration gate is

`research/rank-seven-order-eight-kernel-theorem-verifier.py`

with SHA-256

`334e963b36e88549716e772374c4bf2729b3757b013ebc5482deaf821ab47ee2`

on the verified integration bytes. The repository preserves the complete source lineage in its merge history. From the repository root, run

```
python3 -B research/rank-seven-order-eight-kernel-theorem-verifier.py
python3 -O -B research/rank-seven-order-eight-kernel-theorem-verifier.py
python3 -B research/rank-seven-order-eight-kernel-theorem-verifier.py \
  --print-manifest
```

The quick text-only output has SHA-256

`c9a4b73d64f4aa4ee141541d97897516b58f25019228d359490b321b7f815319`

and the quick manifest-plus-text output has SHA-256

`d153738d36a43d8178598c52de2067453dca1cefce056bf7d911466ce0211ccd.`

The gate executes its own optimized master child, requires byte-identical normal and `-O` output, rejects 21 focused hostile mutations, and checks that 26 proof-critical Python sources contain no optimization-removable `assert`.

Quick mode is an integrity and composition gate: it hashes every child source and mathematical dependency, authenticates the post-XII compressed and raw streams, validates strict schemas and the singleton bytes, and pins receipts from prior exact normal/`-O` child runs. It deliberately does not rerun the expensive semantic children. To do so, run

```
python3 -B research/rank-seven-order-eight-kernel-theorem-verifier.py --full
```

Full mode successfully executes the base bridge, structural replay, packet-chain replay, post-XII replay, exhaustive kernel/census regeneration, all 29,000 direct-rational witness replays, singleton reconstruction, and the imported tricyclic finite gates both normally and under `-O`. It reconstructs the selected 492,812-row census projection in memory and validates the pinned cache field-for-field, but does not write a newly regenerated cache artifact or rerun the three residual historical producer searches. The post-XII normalized proof receipt is pinned as

`1dbd2337043e283e4b8913b89151da6b000cbc516a822f3b5f24bbf94e89eb4b.`

The distinction between quick integrity verification, full read-only replay, and fresh producer recomputation is part of the theorem contract.

Build this manuscript with

```
bash scripts/build-paper.sh rank-seven-order-eight-kernels
```

Scoped nonclaim. Theorem 1.1 concerns only cyclomatic-rank-seven graphs with exactly one positive-rank cyclic block and with suppressed kernel order eight. It makes no claim for kernel orders nine through twelve, for any multiblock rank-seven graph, or for all connected heptacyclic graphs. Its tree-free subfamily was already covered by [2, 3]; this paper's additional scope relative to those results is the uniform treatment of arbitrary rooted-tree attachments within the stated finite kernel frontier. It makes no firstness or unconditional novelty claim. It does not claim fresh reruns of the three residual historical base producer searches. It uses neither parity-changing subdivision monotonicity nor edge-addition monotonicity.

AI disclosure

This manuscript and its proof-object synthesis were generated by an AI coding and mathematical reasoning system from the cited repository proofs and exact certificate artifacts. No human authorship, human review, or human verification is claimed. The result should be assessed from the displayed arguments, explicit trust boundary, frozen sources, and reproducible checks. No external publication was performed as part of this manuscript build.

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[9] *Rank-seven order-eight kernel source and status audit*, 11 November 2026; `research/rank-seven-order-eight-source-status-audit-2026-11-11.md`.

Proof artifacts and reproducibility

[10] Companion manuscript, *Positive Square Energy of Every Tricyclic Graph*, 2026; `all-tricyclic-graphs/paper.tex`. The final gate pins the manuscript, PDF, README, hostile audit, and finite dependency verifiers by SHA-256.

[11] *Rank-seven removable-ear census theorem and exact audit*, 2026; `positive-square-energy/heptacyclic-general/rank-seven-removable-ear-census-theorem.md` and `research/rank-seven-kernel-frontier-census.py`.

[12] *Rank-seven single-block structural theorem: exact reduction and obstructions*, 2026; `positive-square-energy/heptacyclic-general/rank-seven-single-block-structural-theorem.md`.

[13] Exact base-owner precedence bridge and hostile audit, `positive-square-energy/experiments/rank7_order8_typed_precedence_bridge.py`.

[14] Exact read-only replay for structural lanes I–III, `positive-square-energy/experiments/rank7_order8_structural_exact_verifier.py`.

[15] Exact read-only replay for packet lanes IV–XII, `positive-square-energy/experiments/verify_rank7_order8_packet_chain.py`.

[16] *Final order-eight residual: two induced tricyclic territories*, 2026; `positive-square-energy/heptacyclic-general/rank-seven-order-eight-final-two-k4-territory.md`.